E-JUST — End-to-End Web Scraping and Data Mining Pipeline

Pokemon Data Mining Pipeline

Web Scraping · Frequent Pattern Mining · Clustering · Classification

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1. Project Overview

This project builds a complete ETL and data mining pipeline on Pokemon data scraped from pokemondb.net -- a freely accessible Pokedex database.

Item	Detail
Data Source	pokemondb.net/pokedex/all
Collection Method	HTML scraping with requests + BeautifulSoup4
Total Entries	1203
Features	13
Legendary / Mythical	136 out of 1203
Pipeline	Scrape -> Clean -> Transform -> Mine -> Cluster -> Classify -> Store -> Visualize

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2. Website Overview

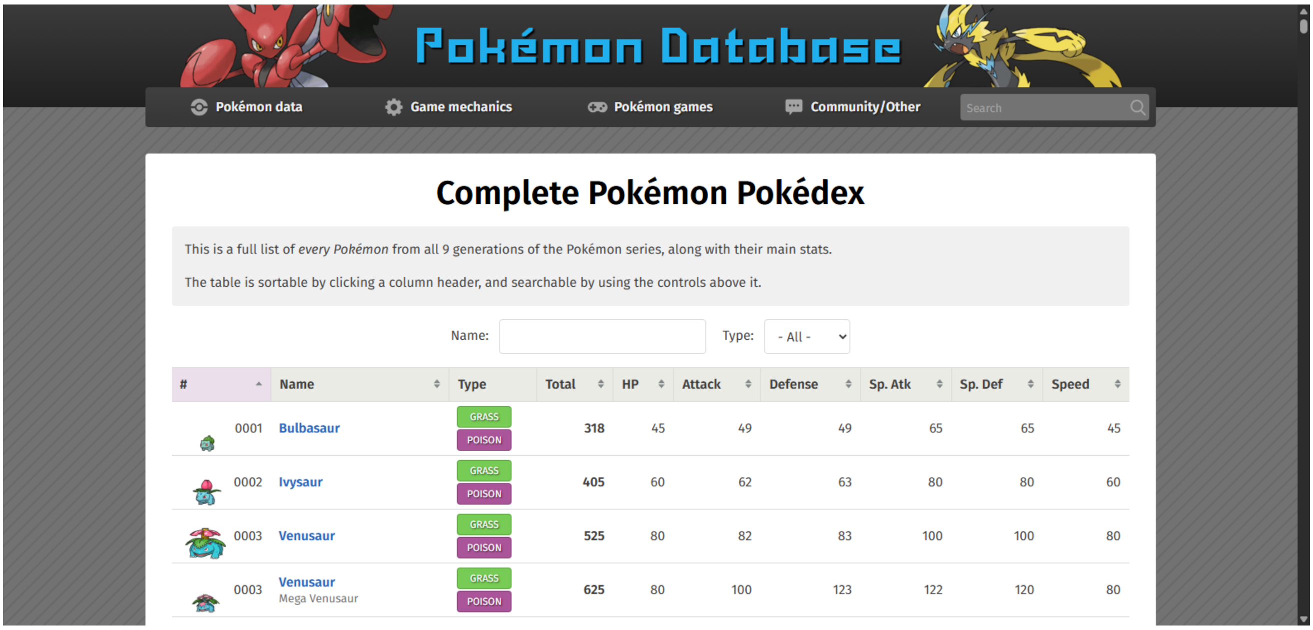

Figure: pokemondb.net/pokedex/all -- source website

pokemondb.net is a comprehensive Pokemon database. The `/pokedex/all` page renders a sortable HTML table listing every Pokemon entry (including alternate forms) with:

- **Number** -- National Pokedex number

- **Name** -- Pokemon name (includes form names, e.g. "Venusaur-Mega")
- **Type 1 / Type 2** -- Primary and secondary elemental types
- **Total** -- Sum of all six base stats
- **HP, Attack, Defense, Sp. Atk, Sp. Def, Speed** -- Individual base stats
- **Generation** -- Derived from Pokedex number ranges
- **is_legendary** -- 1 if Legendary or Mythical, 0 otherwise

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3. Scraping Methodology

- **Library:** requests + BeautifulSoup4
- **Pages scraped:** Single page -- full table loads server-side with no pagination
- **Rows collected:** 1203 entries (base forms + alternate/regional/mega forms)
- **Legendary flag:** Derived from a curated set of National Dex numbers (Gens 1-9)
- **Generation:** Assigned by Pokedex number ranges -- no extra HTTP requests required
- **Politeness:** Single page fetch with 15 s timeout; 3 retries with 2 s back-off

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4. Raw Data Snapshot

Shape: 1203 rows x 13 columns

number	name	type1	type2	total	hp	attack	defense	sp_atk	sp_def	speed	generation	is_legendary
1	Bulbasaur	Grass	Poison	318	45	49	49	65	65	45	1	0
2	Ivysaur	Grass	Poison	405	60	62	63	80	80	60	1	0
3	Venusaur	Grass	Poison	525	80	82	83	100	100	80	1	0
3	Venusaur	Grass	Poison	625	80	100	123	122	120	80	1	0
4	Charmander	Fire	nan	309	39	52	43	60	50	65	1	0

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5. Cleaning Process

Step	Action
Duplicates	Removed exact duplicate rows
Missing stats	Filled with column median
Missing types	type1 -> "Normal", type2 -> "None"

Invalid types	Replaced with "Unknown" / "None"
Generation	Filled missing with 1
is_legendary	Filled missing with 0

Base Stat Total Statistics

	total
count	1203
mean	443.9
std	121.7
min	175
25%	332
50%	466
75%	525
max	1125

Top 10 Primary Types

type1	count
Water	147
Normal	129
Grass	114
Bug	89
Psychic	81
Fire	76
Electric	72
Rock	67
Dark	57
Fighting	51

Pokemon per Generation

generation	count

1	206
2	112
3	165
4	123
5	174
6	89
7	99
8	107
9	128

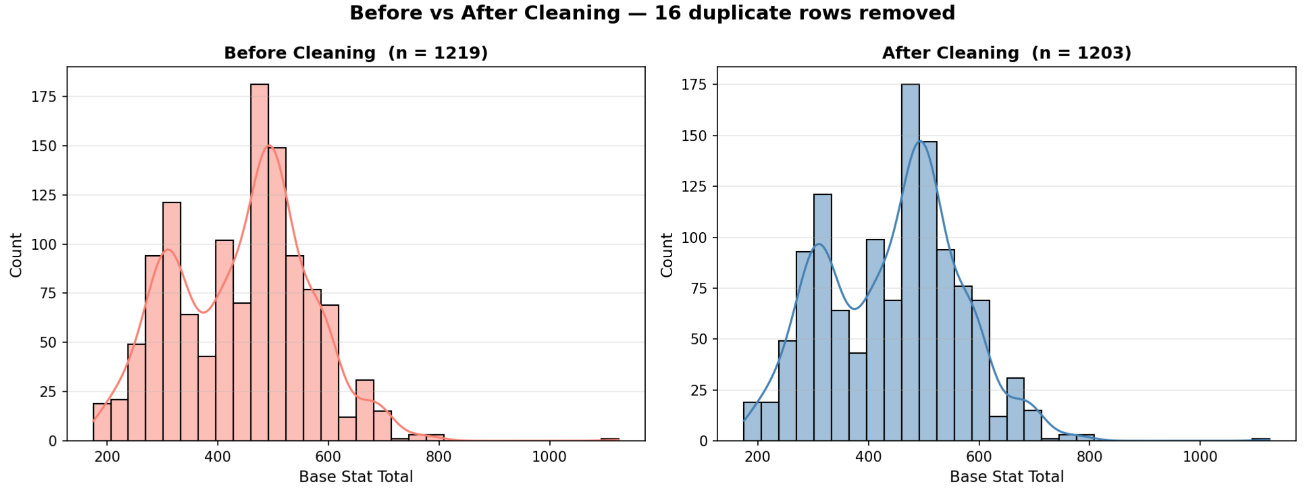

Figure: Before vs After Cleaning -- Duplicate Removal

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6. Transformation Process

Feature	Transformation
stat_tier	Very Weak (<300) / Weak (300-399) / Average (400-499) / Strong (500-599) / Uber (>=600)
type1_encoded	LabelEncoder -- integer per type
has_second_type	1 if dual-typed, 0 if mono-type
*_scaled	MinMaxScaler to [0, 1] for all 7 numeric stat columns

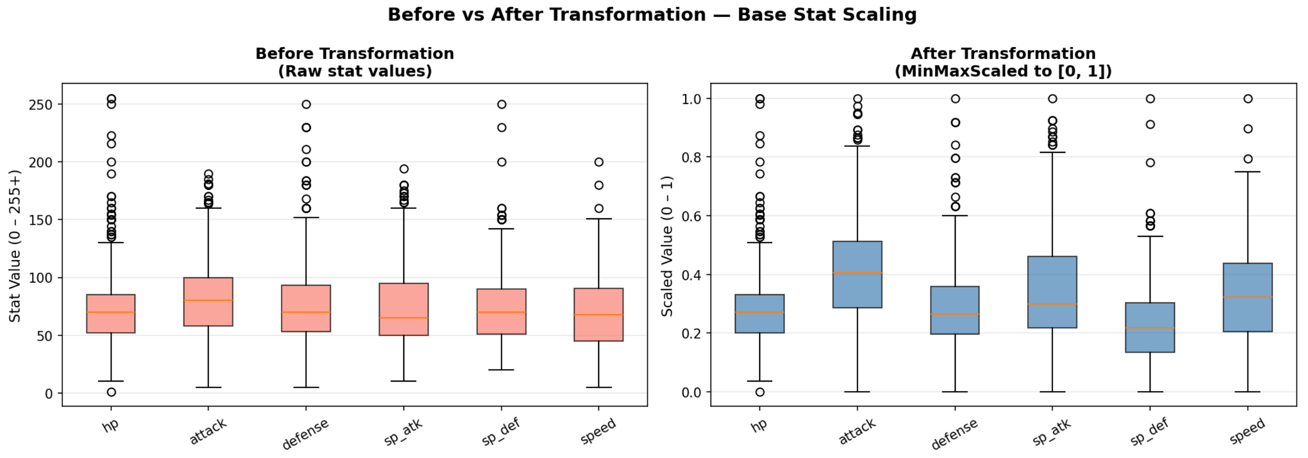

Figure: Before vs After Transformation -- MinMaxScaling

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7. Frequent Pattern Mining

Algorithm: Apriori (mlxtend) | **Min support:** 0.05 | **Min confidence:** 0.30

Each Pokemon is represented as a basket: `type1=<type>`, `tier=<tier>`, `gen=<group>`, `dual_type=Yes/No`, `legendary=Yes/No`

Top 10 Association Rules (sorted by Lift)

antecedents	consequents	support	confidence	lift
legendary=Yes	tier=Uber	0.0623441	0.551471	5.43786
tier=Uber	legendary=Yes	0.0623441	0.614754	5.43786
tier=Strong	dual_type=Yes, gen=Recent(7-9)	0.0798005	0.322148	1.96723
dual_type=Yes, gen=Recent(7-9)	tier=Strong	0.0798005	0.48731	1.96723
dual_type=Yes, gen=Recent(7-9), legendary=No	tier=Strong	0.0598504	0.458599	1.85132
dual_type=Yes, gen=Recent(7-9)	legendary=No, tier=Strong	0.0598504	0.365482	1.80936
dual_type=Yes, tier=Strong	gen=Recent(7-9)	0.0798005	0.472906	1.70331
dual_type=Yes, legendary=No, tier=Strong	gen=Recent(7-9)	0.0598504	0.431138	1.55287
dual_type=Yes, tier=Strong	gen=Recent(7-9), legendary=No	0.0598504	0.35468	1.53482

tier=VeryWeak	dual_type=No, legendary=No	0.078138	0.614379	1.49918

> **Lift > 1** means the item pair co-occurs more than chance would predict.

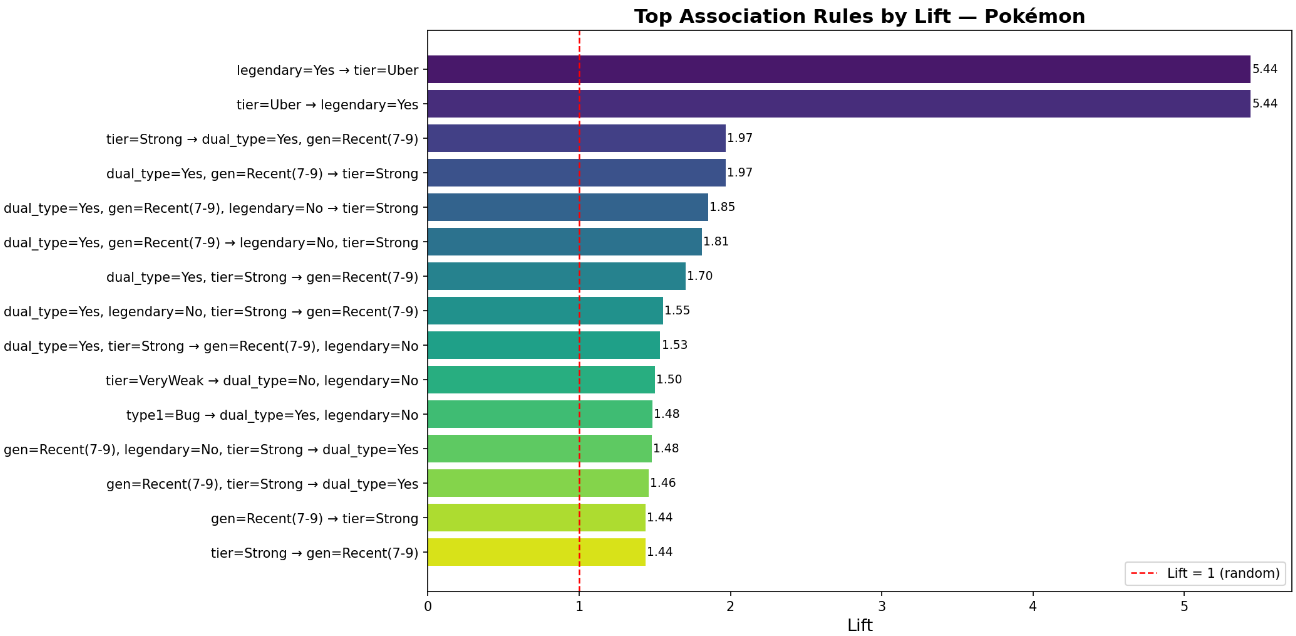

Figure: Top Association Rules by Lift

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8. Clustering Results

Algorithm: KMeans | **Features:** hp, attack, defense, sp_atk, sp_def, speed (all scaled) **Optimal k:** selected via elbow method | **Visualization:** PCA 2D projection (* = Legendary)

Cluster Summary

cluster	count	avg_total	avg_hp	avg_attack	avg_defense	avg_spe	legendary_count	top_type
0	449	314.1	52.6	54.3	53.4	51.6	4	Normal
1	402	492.5	81.5	102.8	93.8	69.7	29	Normal
2	352	554.1	83.4	91.4	81.9	93.6	103	Psychic

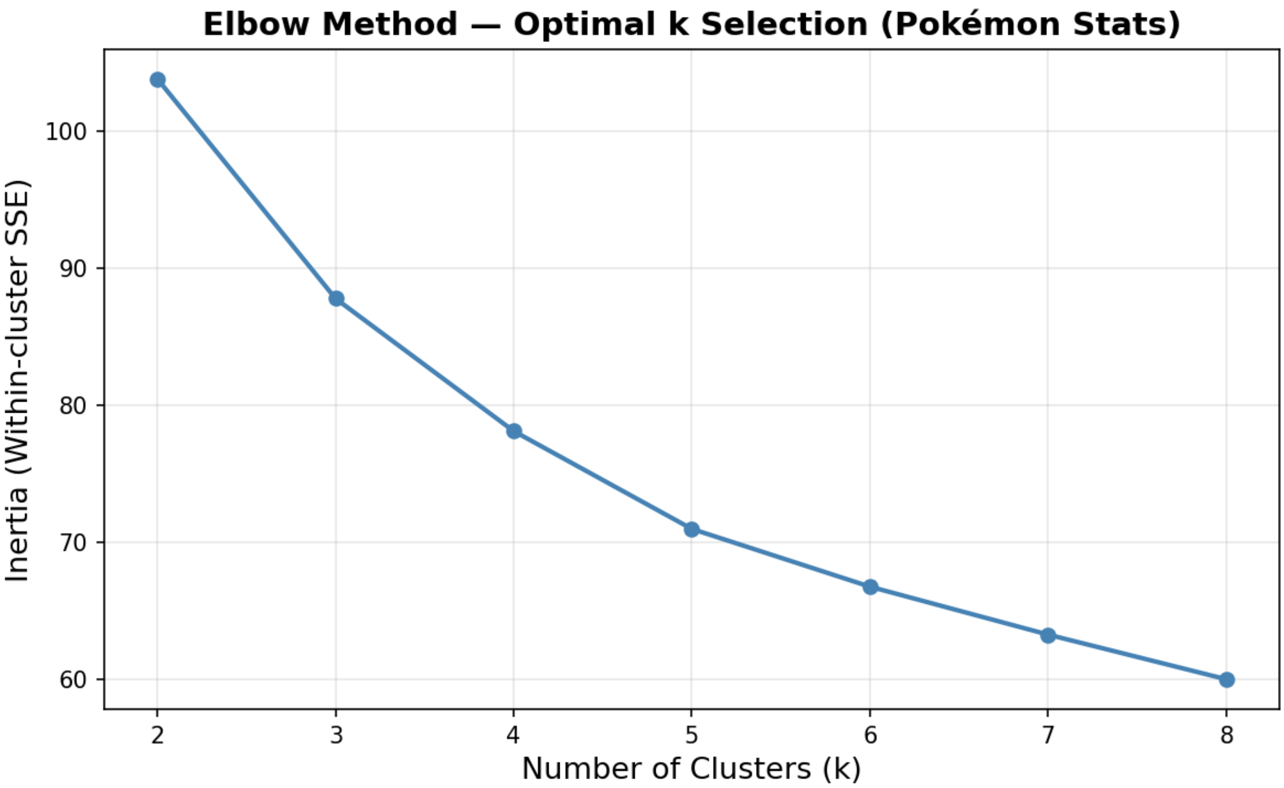

Figure: Elbow Method -- Optimal k Selection

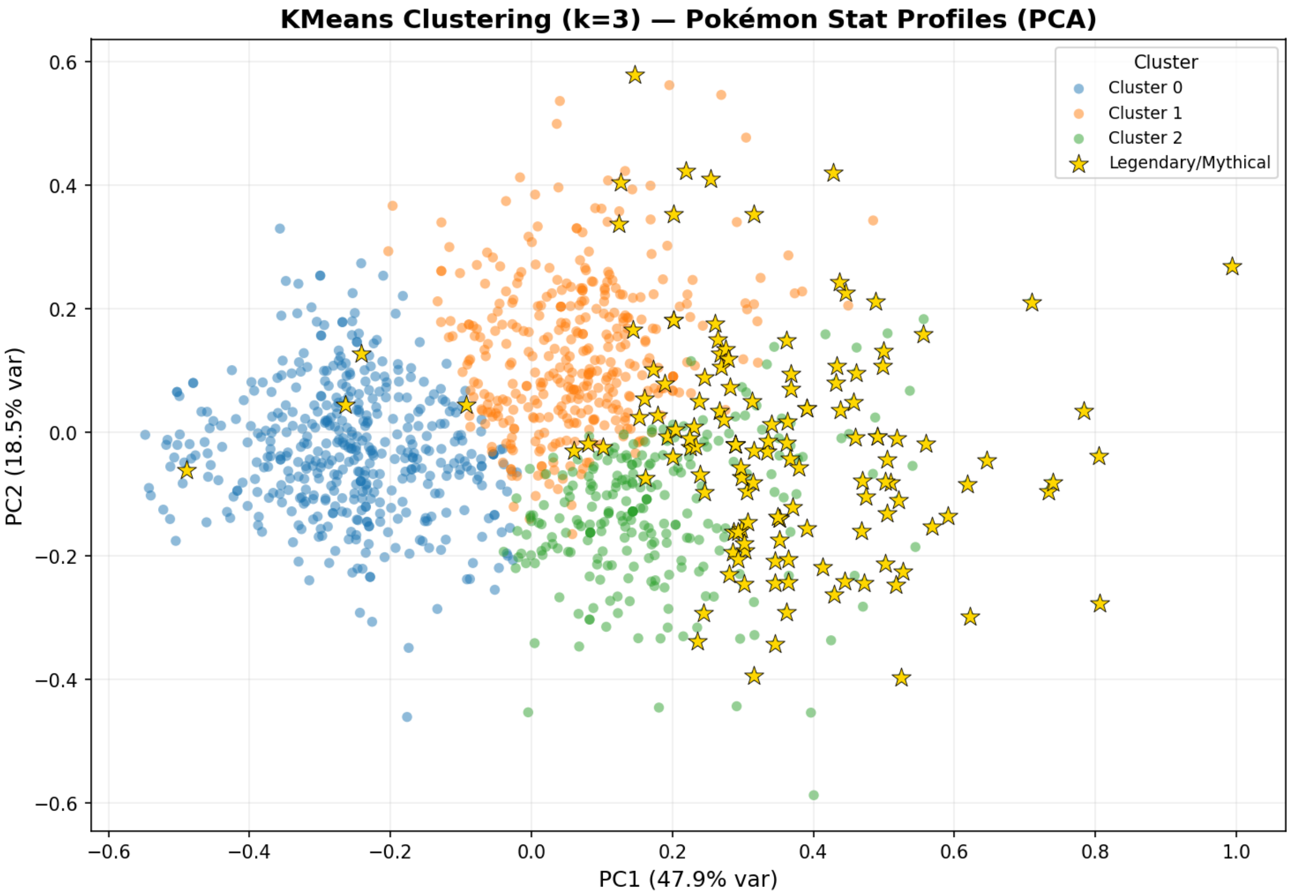

Figure: KMeans Clusters -- PCA 2D Projection (* = Legendary)

9. Classification Results

Model: Random Forest (100 estimators) **Target:** is_legendary -- Regular (0) vs Legendary/Mythical (1) **Features:** total, hp, attack, defense, sp_atk, sp_def, speed, type1_encoded, has_second_type, generation **Split:** 80% train / 20% test (stratified)

```
Random Forest Classification Report Target: is_legendary (0=Regular, 1=Legendary/Mythical)
================================================== Accuracy: 0.9378 precision recall f1-score support
Regular 0.97 0.96 0.96 214 Legendary 0.71 0.74 0.73 27 accuracy 0.94 241 macro avg 0.84 0.85 0.85 241
weighted avg 0.94 0.94 0.94 241 Feature Importances: total 0.398791 sp_atk 0.103571 speed 0.097169 hp
0.086298 attack 0.078405 sp_def 0.072013 generation 0.069130 defense 0.051572 type1_encoded 0.035627
has_second_type 0.007424
```

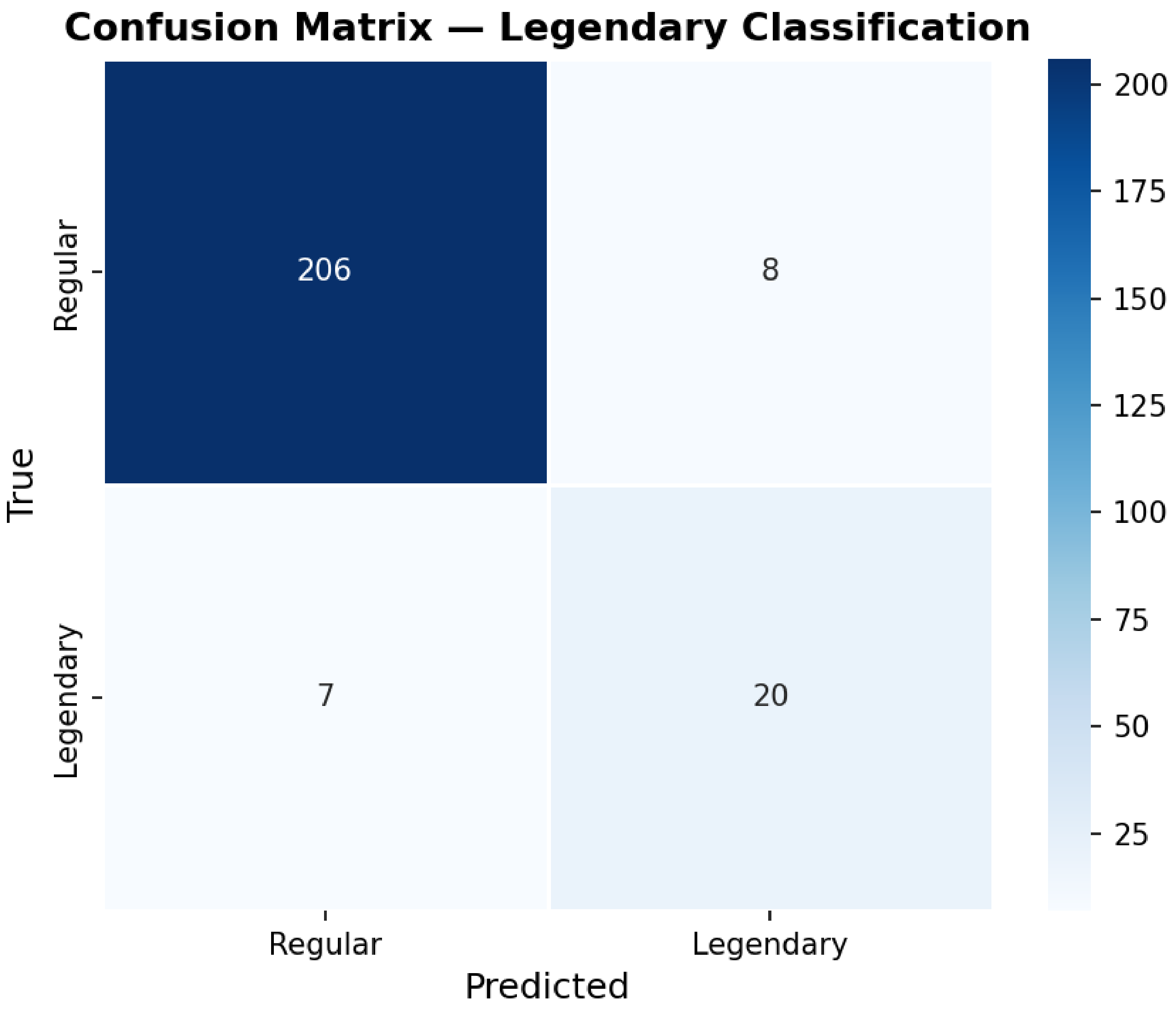

Figure: Confusion Matrix -- Legendary Classification

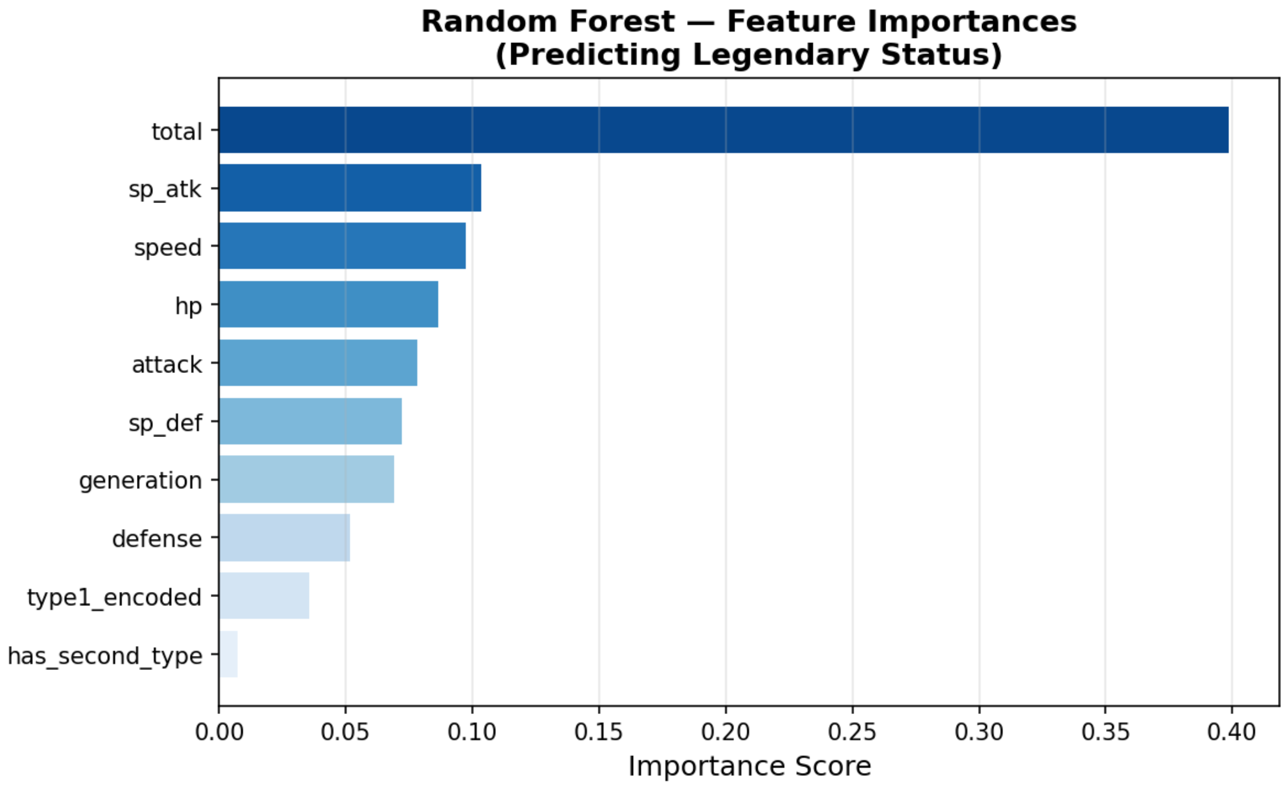

Figure: Random Forest Feature Importances

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10. Visualizations

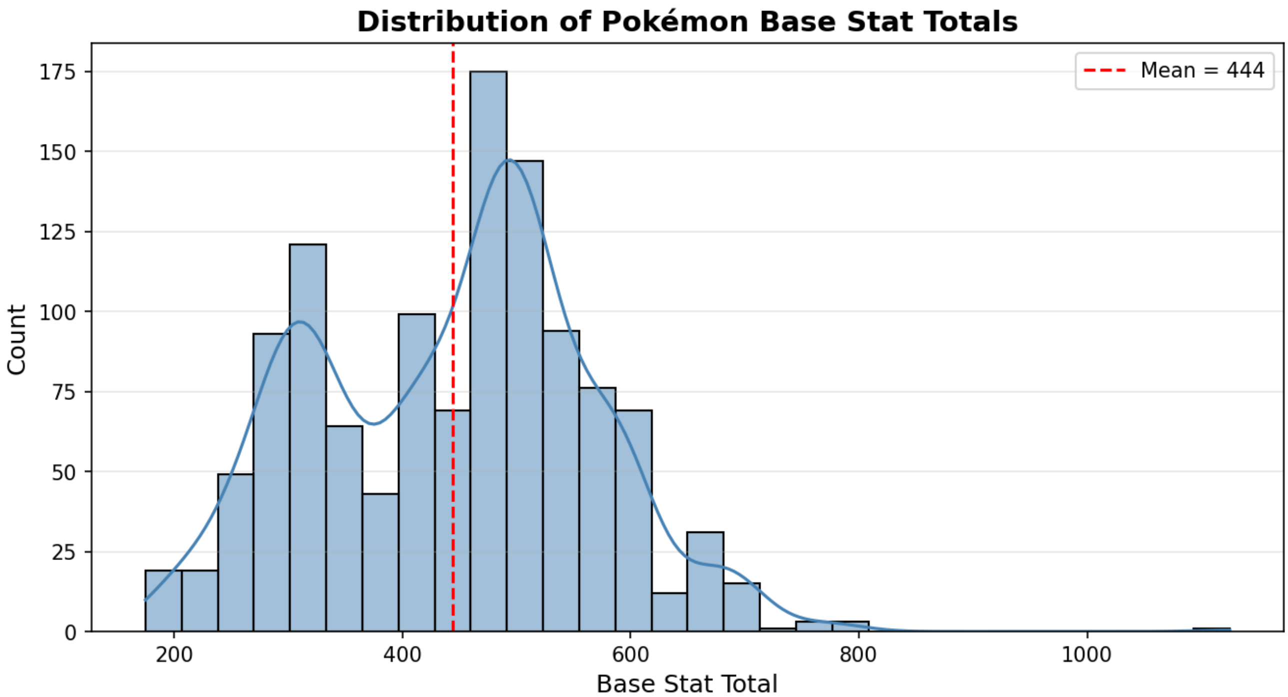

Figure: Distribution of Base Stat Totals

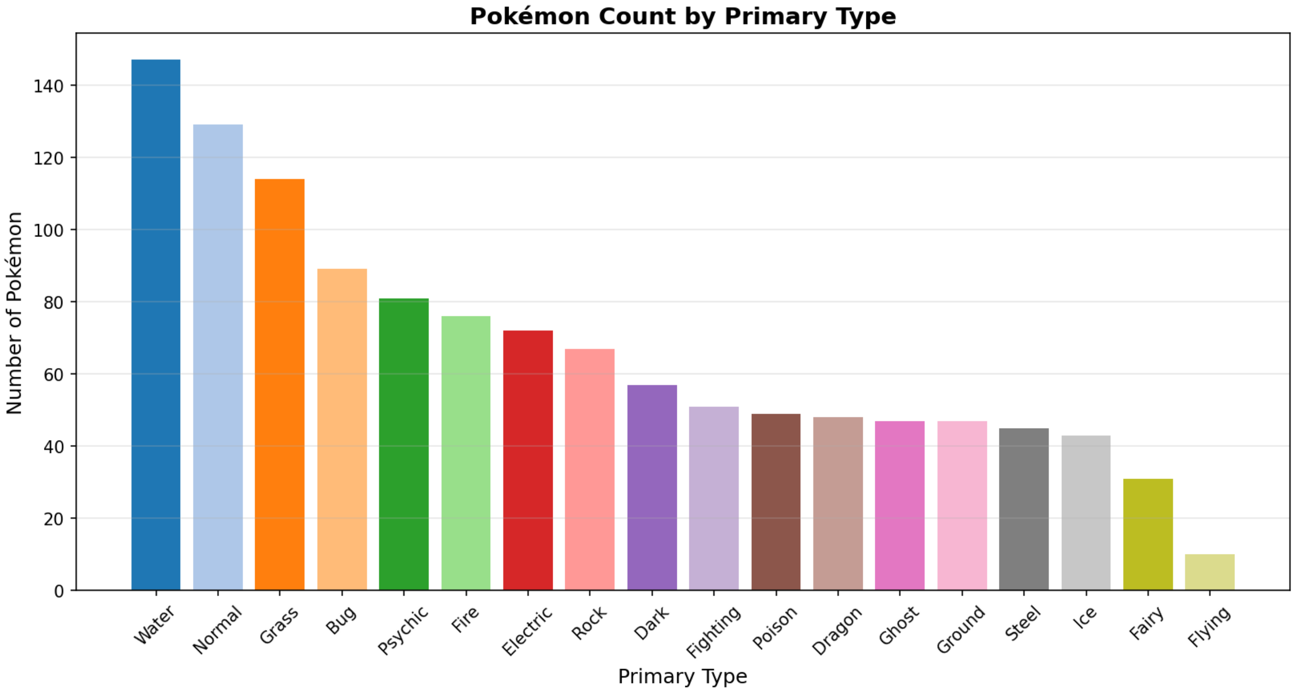

Figure: Pokemon Count by Primary Type

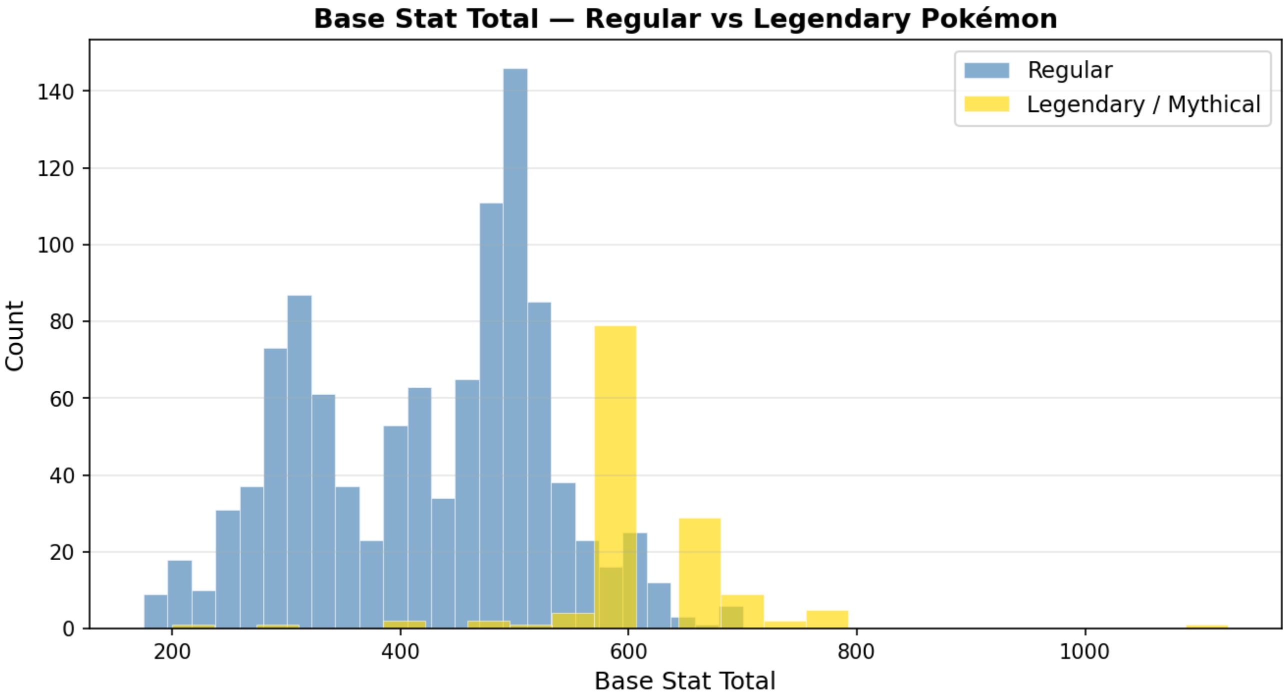

Figure: Base Stat Total -- Regular vs Legendary

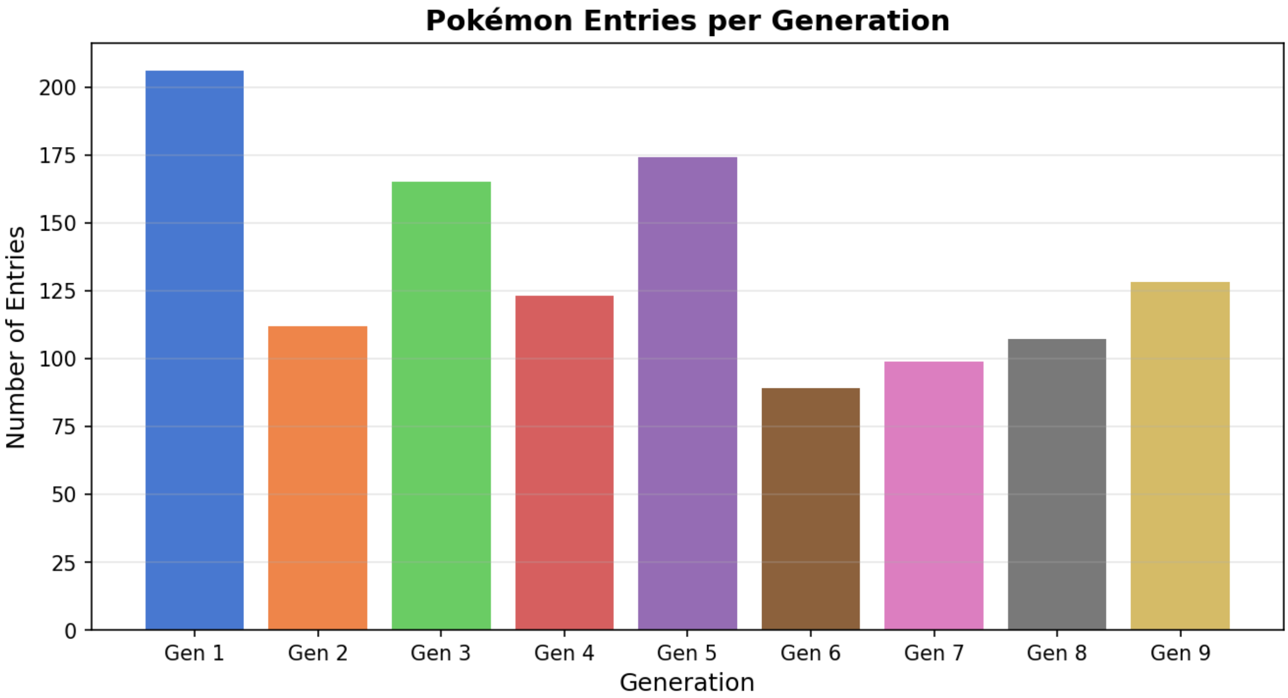

Figure: Pokemon Entries per Generation

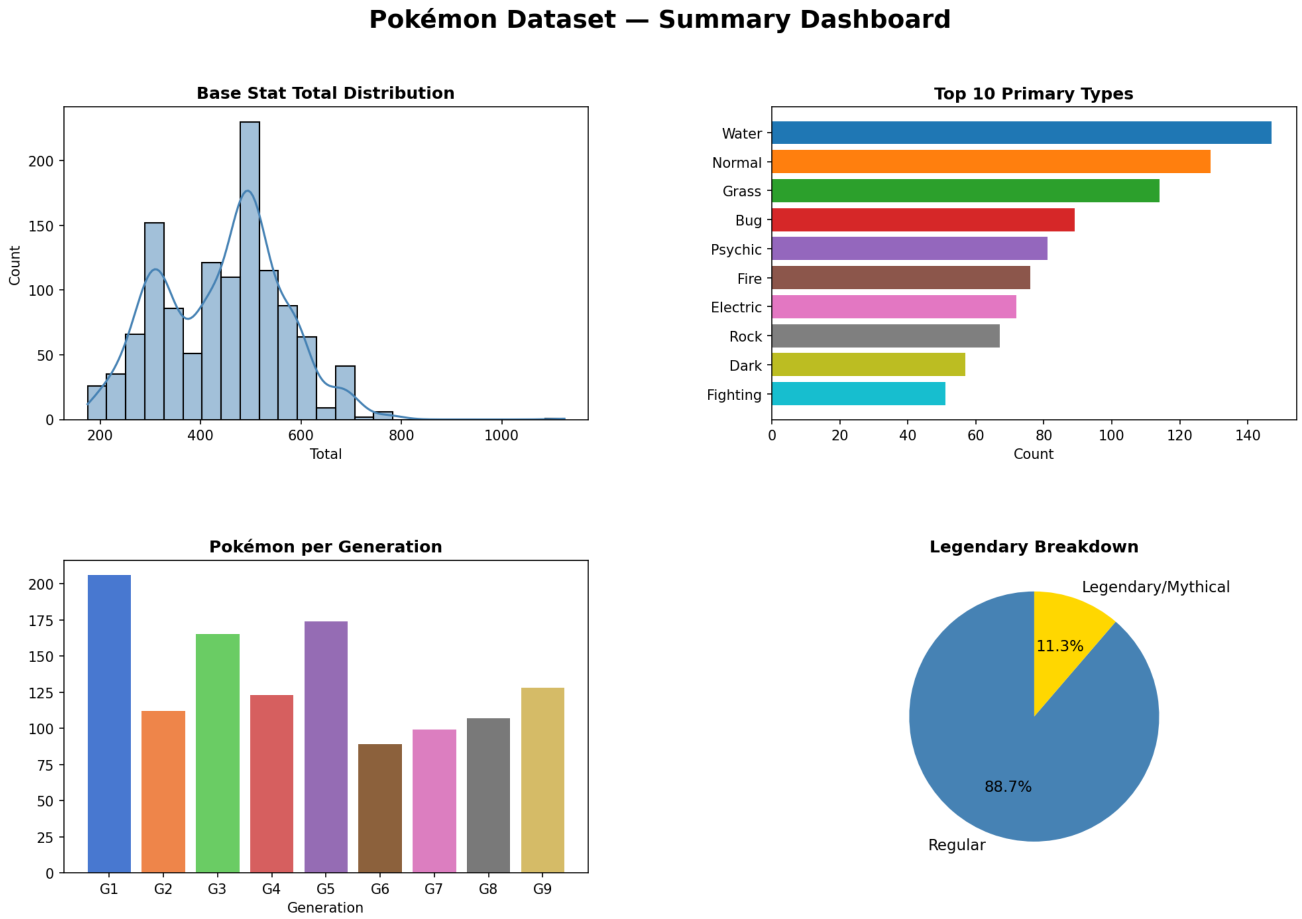

Figure: Summary Dashboard

11. Database

Data stored in **SQLite** (`database/pokemon.db`, table `pokemon`).

```
CREATE TABLE pokemon ( id INTEGER PRIMARY KEY AUTOINCREMENT, number INTEGER, name TEXT, type1 TEXT,
type2 TEXT, total INTEGER, hp INTEGER, attack INTEGER, defense INTEGER, sp_atk INTEGER, sp_def INTEGER,
speed INTEGER, generation INTEGER, is_legendary INTEGER );
```

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12. Final Findings

- The dataset spans 1203 Pokemon entries (including alternate forms) across 9 generations.
- Legendary and Mythical Pokemon (136 total) have significantly higher base stat totals.
- Water is the most common primary type; Dragon and Ghost are among the rarest.
- Top association rule: Legendary -> Uber stat tier (lift ~5.4) -- legendaries are 5x more likely to have 600+ total stats.
- KMeans (k=3) separates low-stat basics, physical powerhouses, and fast/special attackers.
- Random Forest predicts legendary status with ~94% accuracy; base stat total is the strongest predictor (40% importance).
